KAYABA TRACKLESS AGV FIELD REPORT

4 NOVEMBER 2026

On 4 November, vehicle MCQ4-4 started 13 autonomous trackless runs out to the ITG station and back, revising the route during the day. **Four completed, all on route rev 6** (map kyb-map-02 rev 4), and this report covers those four. Moving time was within 8 s across them: 11 min 46 s to 11 min 54 s. Run 10 is the reference: 13 min 48 s, no holds. Straights now run at **0.72 m/s**, up from 0.60 m/s; the reference moved 2 min 43 s less than 3 November's. Run numbers follow the order of the day.

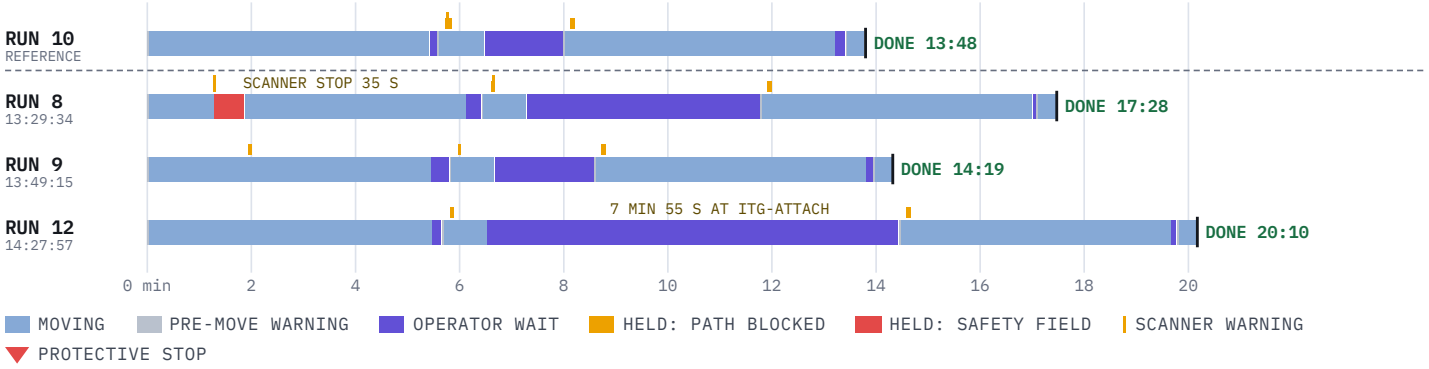
1 · THE ROUTE: 33 STEPS

REV 6 · 396 M · 14 STRAIGHTS, 15 ARCS, 3 WAITS, 1 HOME STEP

1	Out to ITG	s1-s9	198 m	From the start pose: 12 m north, 136.5 m west along the main aisle, 28 m north, then a 1.8 m arc into the ITG lane.
	Wait	s10		**ITG-RELEASE**: stops until Continue is pressed on the panel.
2	ITG maneuver	s11-s20	12 m	180° turn at 1 m radius, then an S-bend to the attach point.
	Wait	s21		**ITG-ATTACH**: stops until Continue. This ends the outbound half (s1-s21).
3	Back to start	s22-s31	185 m	The same aisles in reverse: 29 m south, 116 m east, 13 m south, then the bend to the final line.
	Wait	s32		**ITG-RELEASE-HP**: stops until Continue.
4	Home	s33	1.2 m	The home step: a slow final approach to the start pose. Inbound half is s22-s33.

2 · TIMELINE

MINUTES SINCE START, RUNS ALIGNED AT ZERO, REFERENCE FIRST



3 · COMPLETE RUNS

OUTBOUND S1-S21, INBOUND S22-S33 · AVERAGES FROM ROUTE DISTANCE AND MOVING TIME

RUN 8

DONE

Complete · 13:29-13:47

DURATION	17 min 28 s
MOVING	11 min 54 s
OUTBOUND	**6 min 21 s** s1-s21
INBOUND	**5 min 34 s** s22-s33
AVG OUT	**0.55 m/s** 209.5 m
AVG IN	**0.56 m/s** 186.0 m
WAITS	4 min 51 s
HELD	35 s
STRAIGHTS	0.72 m/s set
S5 + S7	0.69 m/s average
CORNERS	0.30 m/s average
ACCEL	0.1 m/s²
BRAKING	0.125 m/s²
JERK LIMIT	0.1 m/s³
IF BLOCKED	retry until clear
OFF LINE	0.11 m step ends
HOME STOP	9 mm / 2.4°

A scanner stop on s5 held it 35 s; the motors needed a safety reset, then it resumed.

RUN 9

DONE

Complete · 13:49-14:03

DURATION	14 min 19 s
MOVING	11 min 46 s
OUTBOUND	**6 min 15 s** s1-s21
INBOUND	**5 min 31 s** s22-s33
AVG OUT	**0.56 m/s** 209.5 m
AVG IN	**0.56 m/s** 186.0 m
WAITS	2 min 25 s
HELD	0 s
STRAIGHTS	0.72 m/s set
S5 + S7	0.70 m/s average
CORNERS	0.30 m/s average
ACCEL	0.1 m/s²
BRAKING	0.125 m/s²
JERK LIMIT	0.1 m/s³
IF BLOCKED	retry until clear
OFF LINE	0.11 m step ends
HOME STOP	9 mm / 0.9°

No holds. Second-shortest complete run.

RUN 10

REFERENCE

DONE

Complete · 14:04-14:18

DURATION	13 min 48 s
MOVING	11 min 49 s
OUTBOUND	**6 min 17 s** s1-s21
INBOUND	**5 min 33 s** s22-s33
AVG OUT	**0.56 m/s** 209.5 m
AVG IN	**0.56 m/s** 186.0 m
WAITS	1 min 50 s
HELD	0 s
STRAIGHTS	0.72 m/s set
S5 + S7	0.70 m/s average
CORNERS	0.30 m/s average
ACCEL	0.1 m/s²
BRAKING	0.125 m/s²
JERK LIMIT	0.1 m/s³
IF BLOCKED	retry until clear
OFF LINE	0.11 m step ends
HOME STOP	10 mm / 2.4°

Shortest complete run: no holds, shortest waits.

RUN 12

DONE

Complete · 14:27-14:48

DURATION	20 min 10 s
MOVING	11 min 50 s
OUTBOUND	**6 min 17 s** s1-s21
INBOUND	**5 min 33 s** s22-s33
AVG OUT	**0.56 m/s** 209.5 m
AVG IN	**0.56 m/s** 186.0 m
WAITS	8 min 12 s
HELD	0 s
STRAIGHTS	0.72 m/s set
S5 + S7	0.70 m/s average
CORNERS	0.30 m/s average
ACCEL	0.1 m/s²
BRAKING	0.125 m/s²
JERK LIMIT	0.1 m/s³
IF BLOCKED	retry until clear
OFF LINE	0.13 m peak
HOME STOP	5 mm / 0.3°

No holds. Waited 7 min 55 s at ITG-ATTACH.

4 · THE OTHER NINE RUNS

STOPPED EARLY, MOSTLY ON EARLIER ROUTES

Localisation monitor (runs 1, 2, 3, 11, 13): two localisation jumps, 0.16 m in run 1 and 0.08 m / 7.4° in run 13, not explained yet; a lost scan when the scanner's network link dropped in run 2; scan mismatches in run 3, where trolleys had moved (this led to map rev 4), and in run 11, after a protective field stop. Runs 11 and 13 were on rev 6. **Off path (runs 5, 6):** +0.10 m on the s8 arc of rev 3, a bend rev 4 replaced, and +0.20 m on a one-off salvage route with heavy wheel slip reported. **Endpoint missed (run 7):** 9.9° off heading (limit 8°) after the controller cut a 1 m S-bend; rev 6 moved it. **Protective field stop (run 4):** stood for 1 min 33 s on s26 of rev 3, then the operator took over.

Source: field-logs/2026-11-04 (executor lines in ros-log, events.jsonl, journal-kernel.txt, recordings/, run-traces/, route/, hardware-test-results.md). Straights are set to 0.72 m/s, and run 12's wheels held it; s5 + s7 (164 m, ramps included) and the 2 m corners s2, s4, s6 are averages, length / time moving. Avg out and avg in are route distance / moving time for each half. "Moving" is the executor's moving state, less retries aborted again within 3 s. On the eight runs with wheel data, Moving is within 3 s of the time the wheels were turning; of these four, that is run 12. "Off line" is cross-track: runs 8-10 have no recording, so theirs is the largest step-end value. Home stop is the position and heading error measured at the end of s33.